Aviation Accidents & Flight Safety Intelligence

An explanatory data story detailing U.S. and global flight safety metrics, structural durability, weather impact, and mechanical engineering redundancy (1982-2026)

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GitHub Repository: Aviation-Safety-Data-Visualization

Project Overview & Data Sourcing

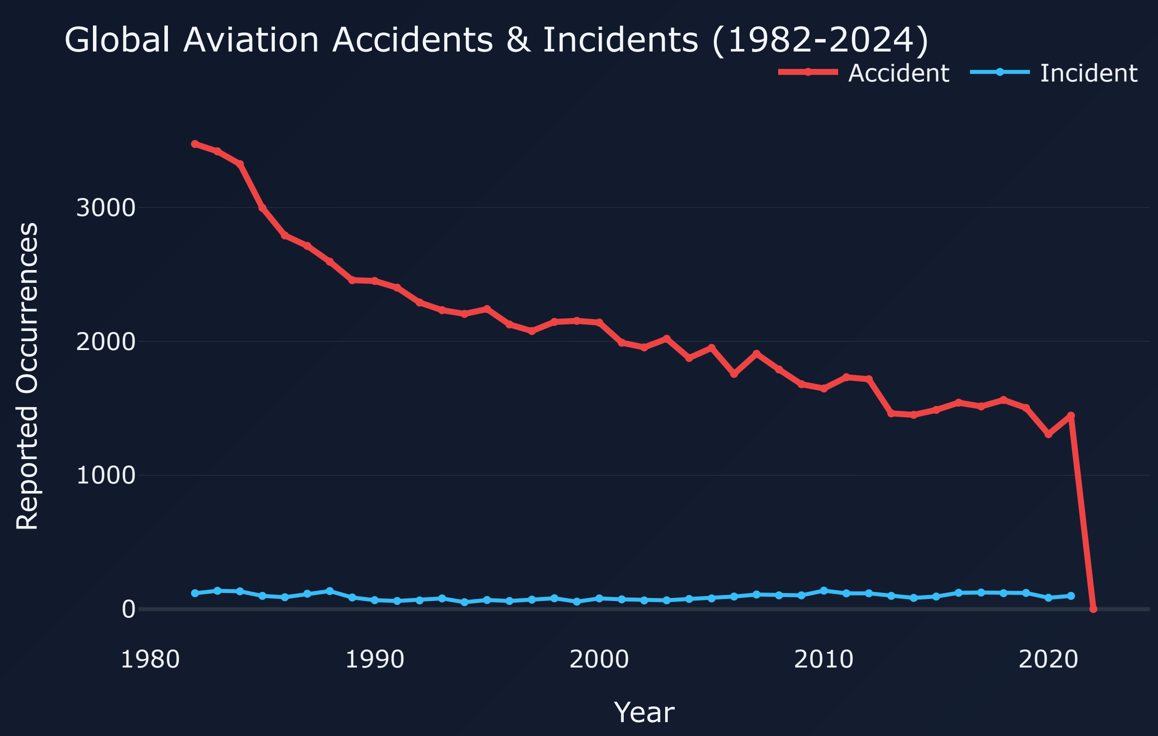
Investigating civil flight safety using national databases

The Goal: To determine flight safety risks across various aircraft dimensions, assessing weather hazards, engine types, and structural integrity parameters.

The Dataset: The National Transportation Safety Board (NTSB) database covering aviation accidents from 1982 to 2026. Includes structural damage severity, geocoded accident coordinates, and engine redundancies.

Accidents Continue to Decline

Annual occurrence trends since 1982 (Q1 Analysis)

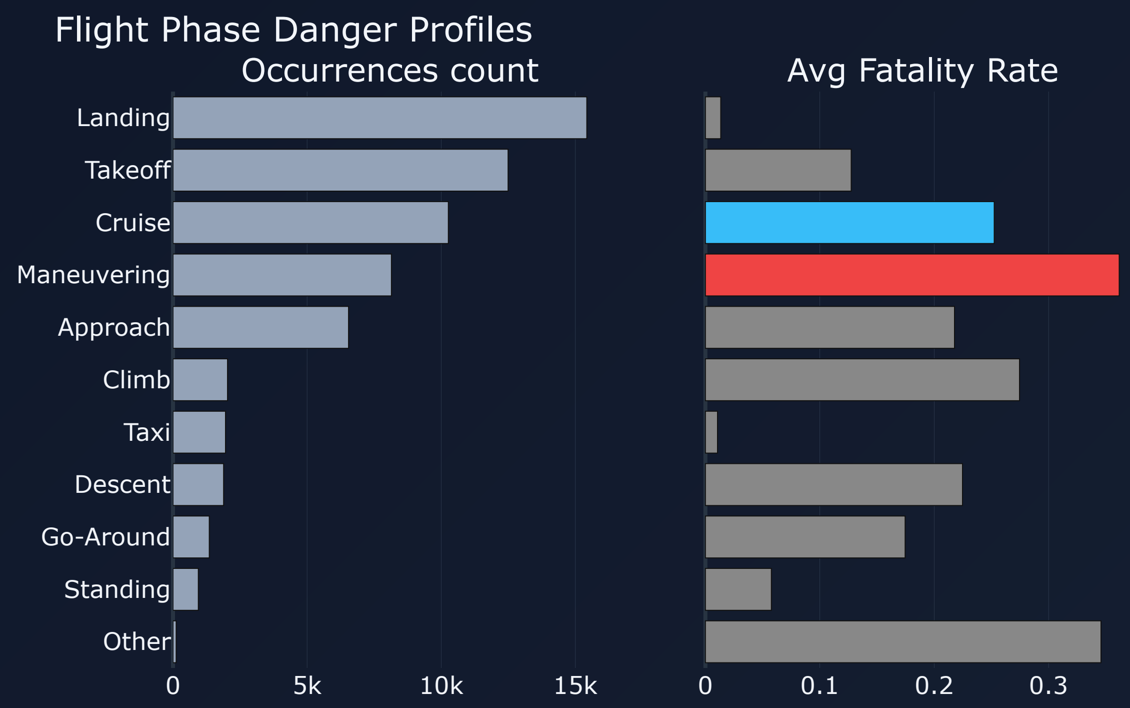


Engineering Triumphs: Global accidents have dropped consistently, highlighting successes in avionics, flight-envelope protections, and pilot training systems.

Incident Volatility: Minor "Incidents" show a flatter trend, likely due to more stringent logging guidelines over time.

Flight Phase Risk Profiles

When do incidents happen, and when are they fatal? (Q2 Analysis)

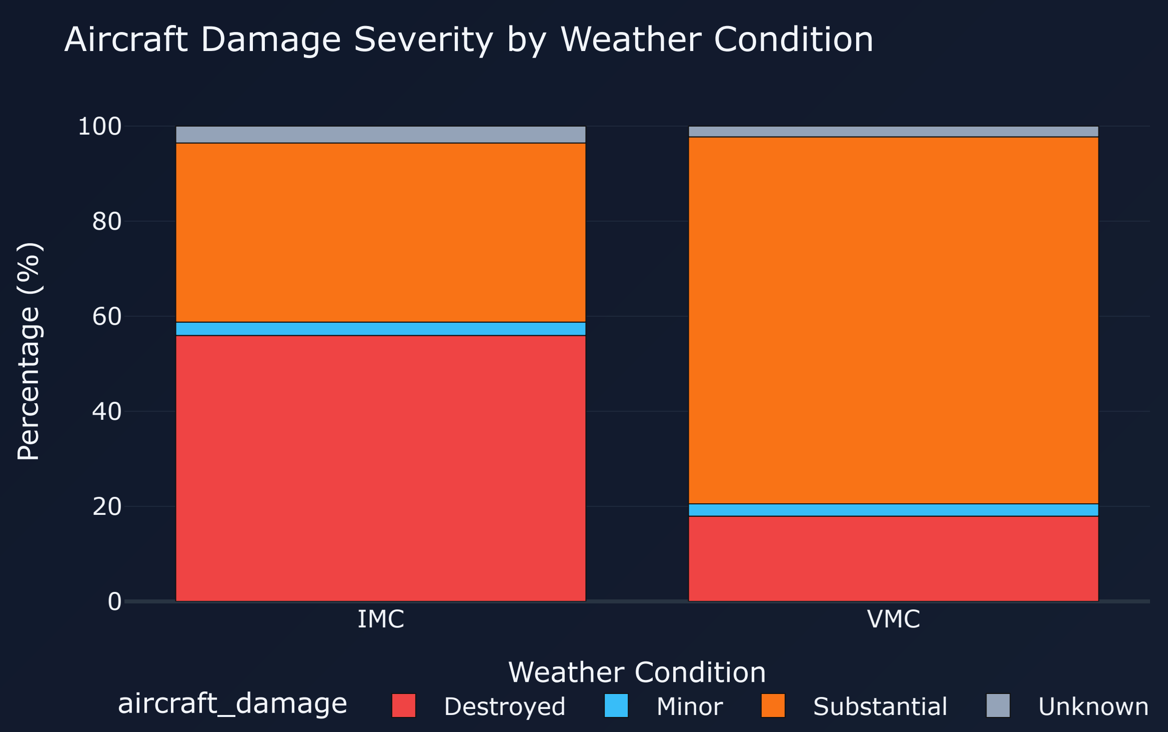


High-Activity Exposure: Landing and Takeoff constitute the vast majority of incidents because of complex aerodynamics and high pilot workload.

Cruise Lethality: Cruise and Maneuvering are statistically the most lethal phases; an emergency in these phases often involves higher speeds and altitudes, leading to greater impact velocities.

Instrument Weather Multiplies Damage

Weather conditions vs. aircraft hull loss (Q3 Analysis)

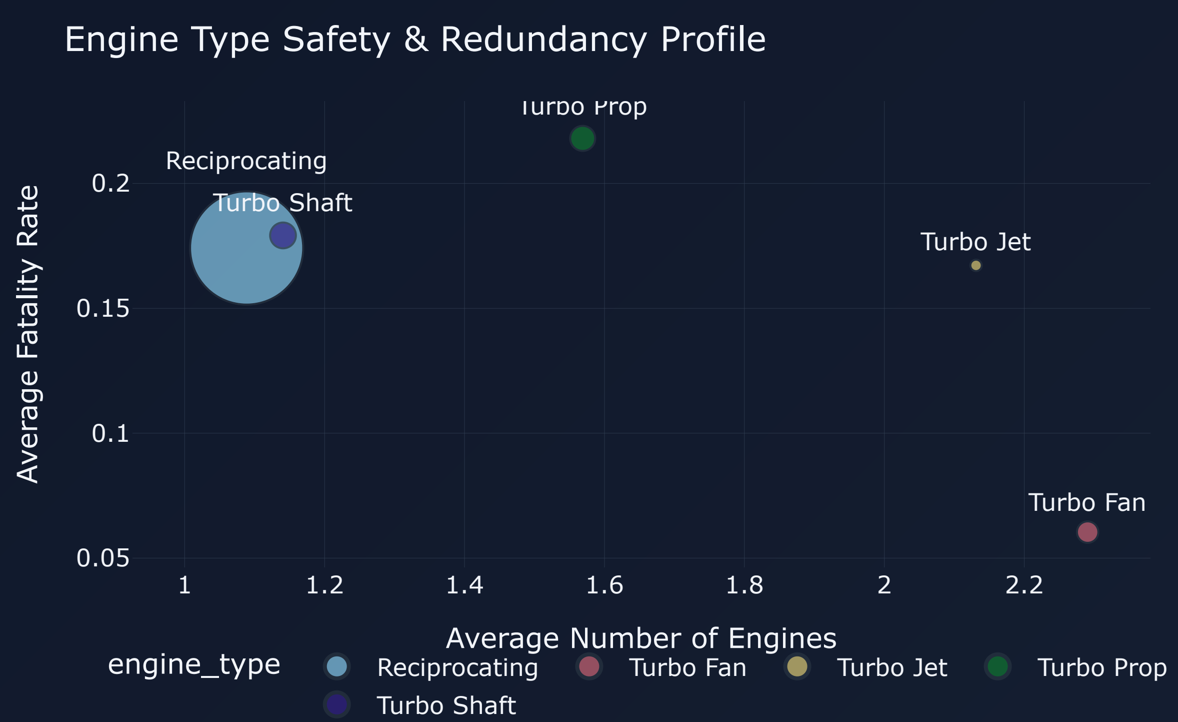


Visibility Hazards: Instrument Meteorological Conditions (IMC) lead to completely destroyed aircraft in nearly 50% of incident records.

Visual Flights: Under Visual Conditions (VMC), pilots maintain a visual reference with the horizon, keeping the overwhelming share of minor or substantial damage profiles.

Engine Configurations & Fatality

Engine architectures and passenger safety (Q4 Analysis)

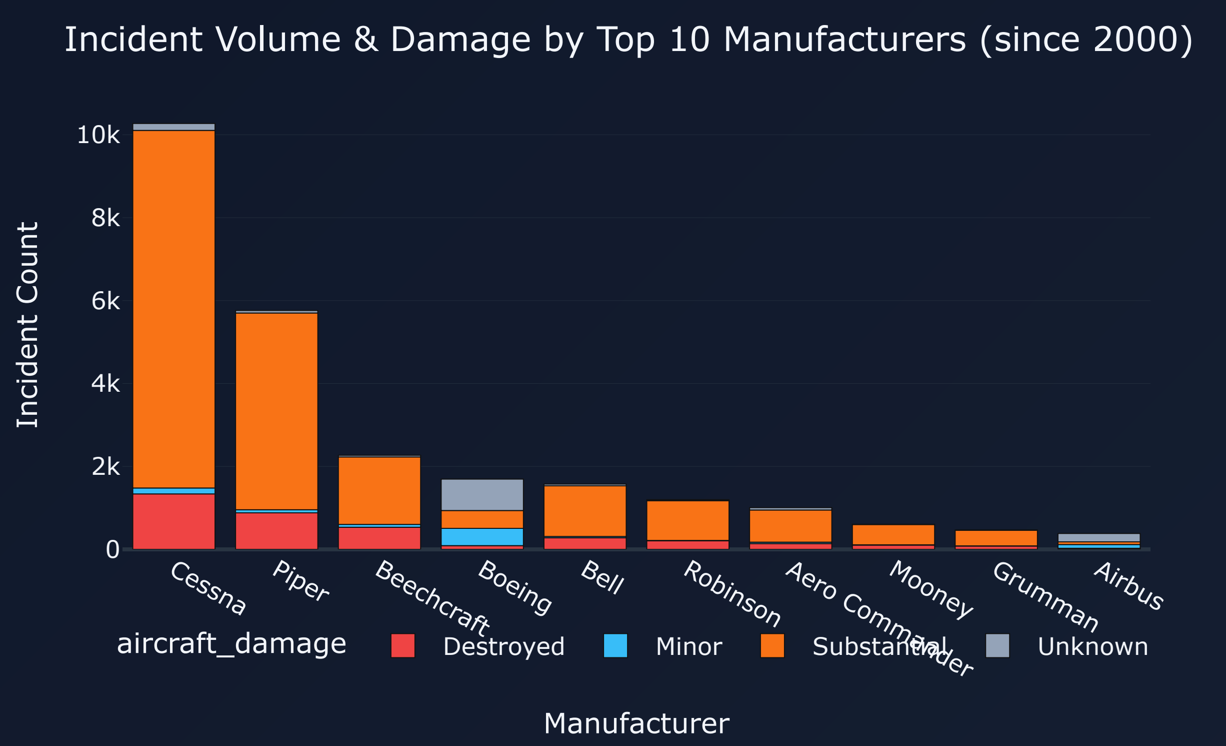


Reciprocating Props: Carry the highest average fatality rates, as they are typically fitted to single-engine recreational light aircraft without backup systems.

Commercial Jets: Turbofan/Turbojet engines have the lowest average fatality rates due to multi-engine redundancies and rigorous commercial airline maintenance logs.

Incident Volumes by Manufacturer

Top 10 brands and their aircraft damage records (Q5 Analysis)

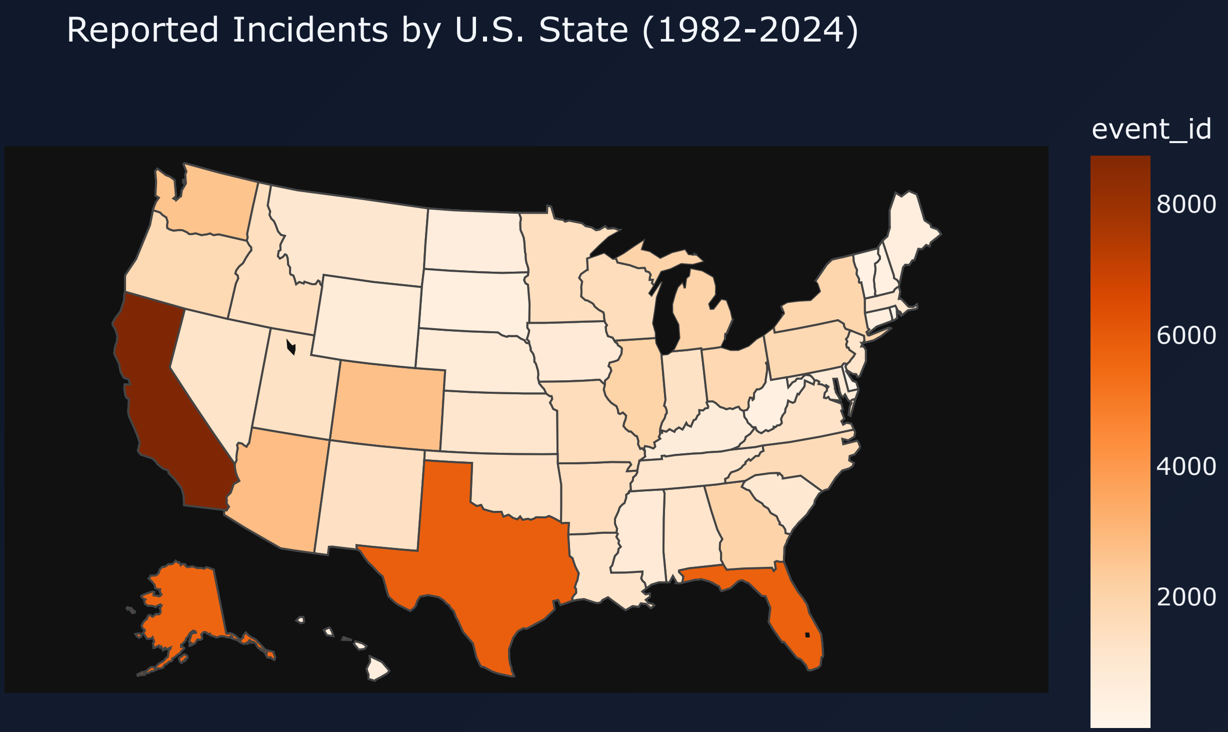


General Aviation Exposure: Cessna and Piper dominate absolute volume due to their vast fleet presence in flight training schools and recreational operations.

Commercial Builders: Boeing shows a lower incident footprint in terms of major hull damage, reflecting commercial airline fleet oversight.

United States Incident Hotspots

State-level incident distribution map (Q6 Analysis)

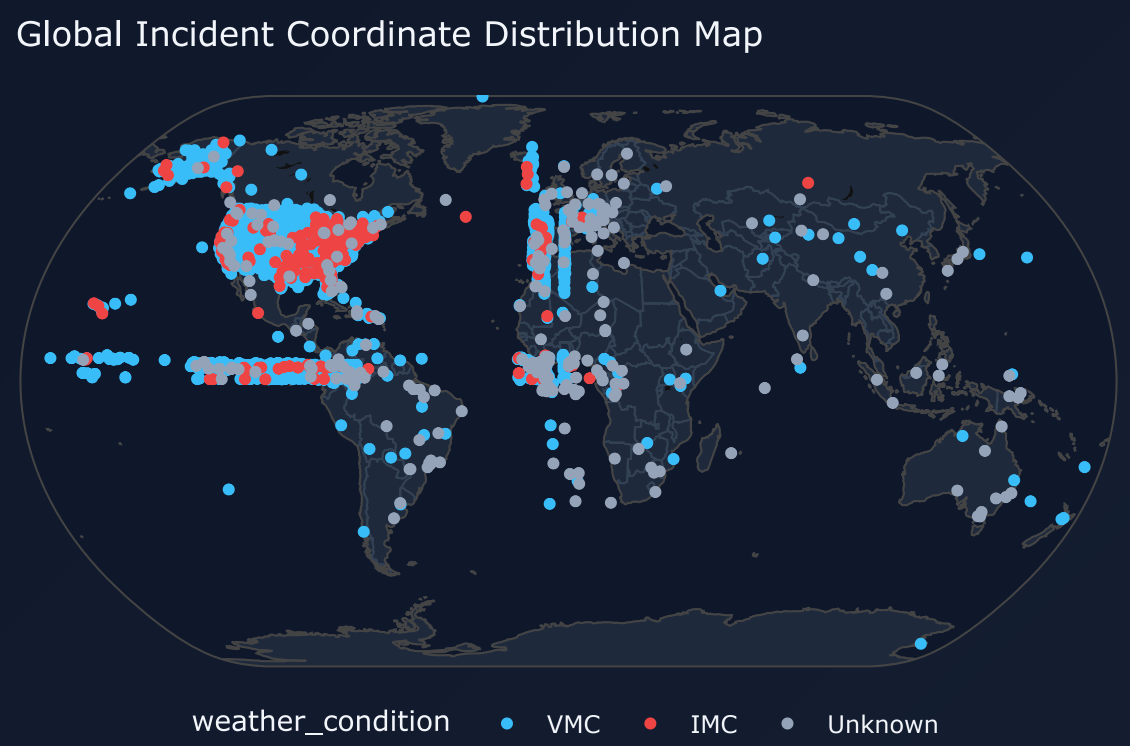


The High-Traffic States: California, Texas, and Florida host the most incidents. This correlates directly with high general aviation traffic and private pilot certifications.

Training Hubs: These states feature favorable weather year-round, making them hubs for private pilot academies.

Global Incident Distribution

Geographic hotspots of aviation incidents globally (Q7 Analysis)

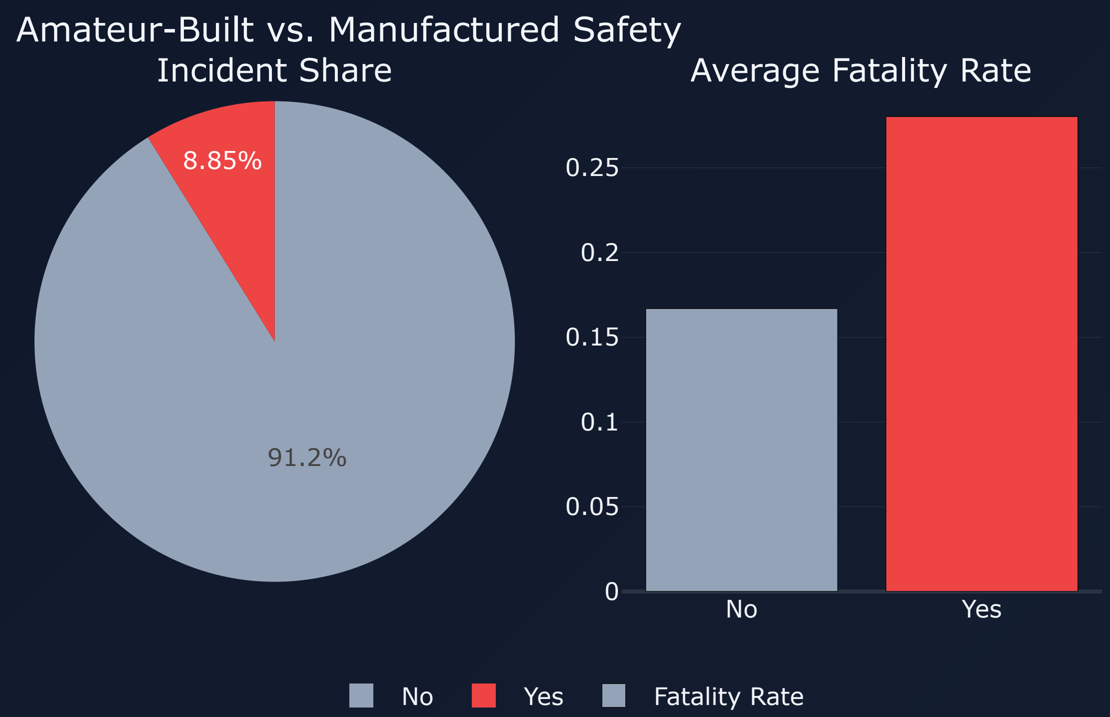


Global Reporting Hubs: The majority of incidents are tracked in North America due to dense aviation infrastructure and NTSB reporting mandates.

Transoceanic Flights: Outlying geocoded coordinate clusters match major oceanic flight corridors and international shipping routes.

Amateur-Built vs. Manufactured Safety

Comparing safety metrics of homebuilt and production models (Q8 Analysis)

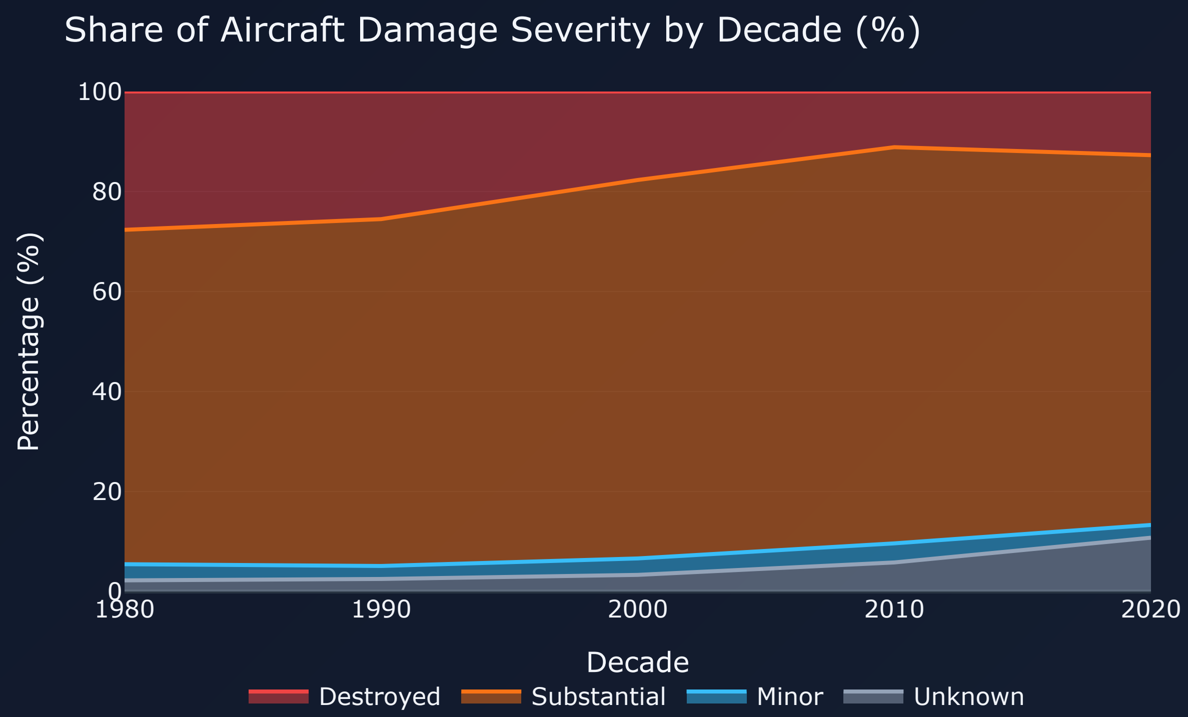


Incident Volume: Homebuilt (amateur-built) aircraft represent less than 10% of total reported incidents in the database.

Fatality Disparity: However, homebuilt aircraft carry a **50% higher average fatality rate** when an accident does occur, highlighting private manufacturing vulnerabilities.

Structural Integrity Progress

Hull loss rates by decade (Q9 Analysis)

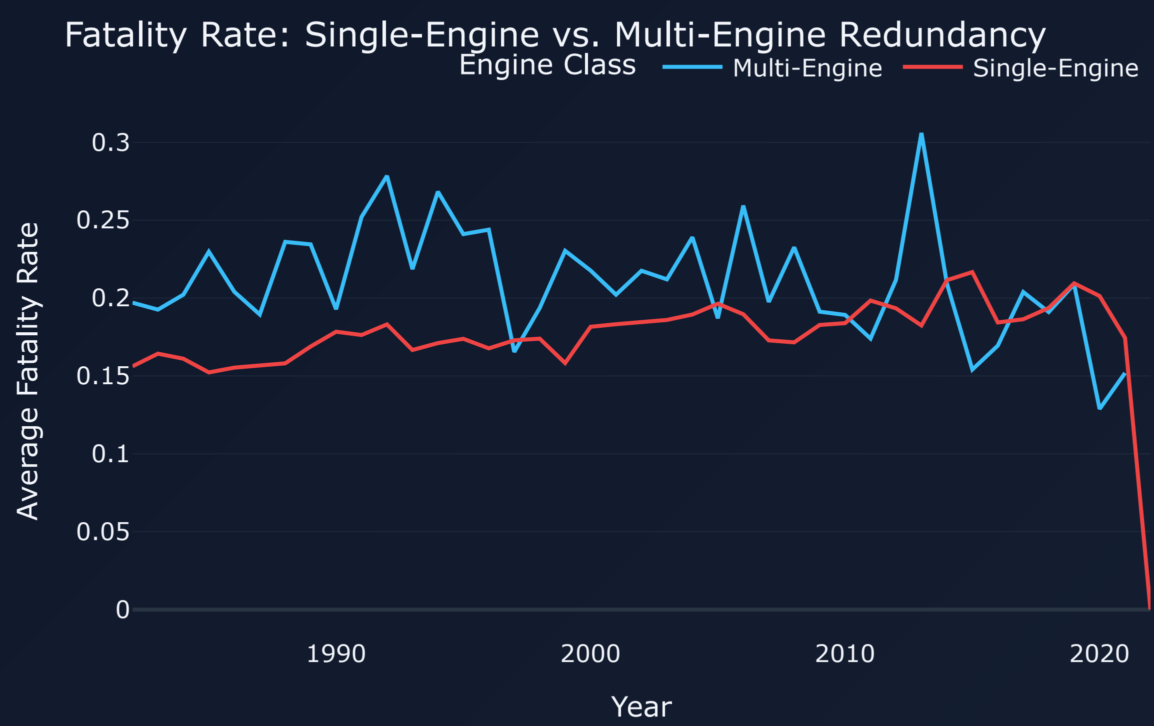


Halving Hull Loss: The percentage of crashes resulting in a "Destroyed" aircraft has shrunk consistently from the 1980s to the 2020s.

Resilient Cabins: Improvements in composite materials, cabin structural cages, and landing gear energy absorption have dramatically increased crash survival rates.

The Power of Engine Redundancy

Single-Engine vs. Multi-Engine fatality trends (Q10 Analysis)

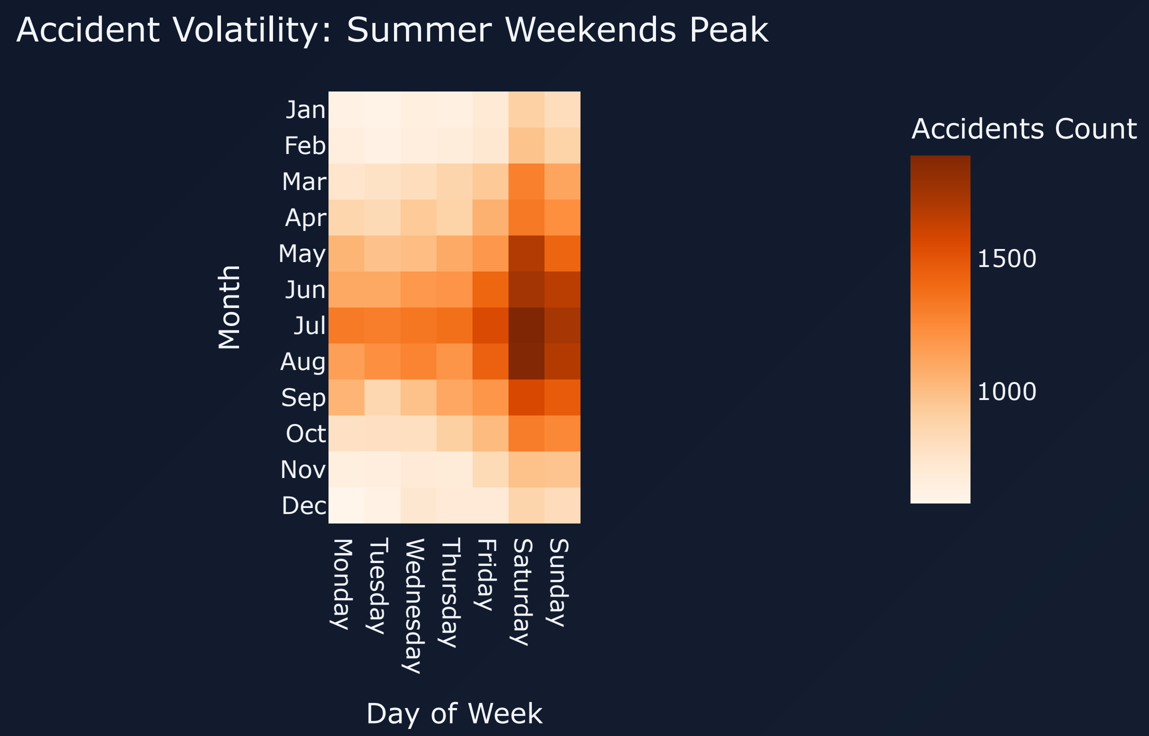


Redundancy Decouples Failure: Multi-engine planes have a fatality rate that is consistently lower than single-engine planes.

The backup engine: If one engine fails on a twin-engine jet, the aircraft can safely fly to an emergency airport using the remaining engine.

Seasonal & Operational Patterns

Investigating accident rates by Month and Day of Week (Q11 Analysis)



The Summer Peak: Accident volumes spike significantly in June, July, and August. This corresponds with recreational and private flying schedules in summer weather.

Weekend Volatility: Saturday and Sunday show elevated incident footprints across all months, aligning with flight school operations and private pilot leisure flights.

Conclusions & Takeaways

1. Engineering saves lives: Decades of updates to aircraft hulls, fire retardant cabin interiors, and engine failure redundancy have made flying progressively safer.

2. General Aviation needs focus: Cessna/Piper single-engine prop aircraft represent the largest share of accidents and fatality rates, indicating a need for better pilot warnings and automation in light aircraft.

3. Weather risk remains constant: Instrument weather (IMC) continues to multiply aircraft damage, proving that low visibility is the biggest threat to manual flight operations.

Quick Start & Navigation Guide

Instructions for professors, reviewers, and developers

🖥️ How to Run the Project

Streamlit Dashboard: Run `.venv/bin/python3 -m streamlit run app.py` to launch the interactive app.

Jupyter Notebook: Open `analysis.ipynb` inside Anaconda or Jupyter to see the full code and data query outputs.

Re-run Pipeline: Execute `python download_and_clean.py` to download and rebuild the datasets.

📊 Presentation Navigation

Widescreen Scrolling: Scroll vertically to browse through the 15 slides. Each slide is sized to fit exactly 100vh.

Interactive Charts: All charts in this deck are live Plotly graphs. Hover to see exact counts, drag to zoom, and double-click to reset.

Print to PDF: Press `Cmd/Ctrl + P`, select **Landscape**, set **Margins: None**, enable **Background graphics** to generate a clean PDF deck.